CS24353 — ELECTIVE (3.0 CR) | MODULE IV NOTES | BIT MESRA | NEP SCHEME

Module 4: Verification, Validation & Software Testing

Black-Box (ECP/BVA), White-Box Cyclomatic Complexity V(G), V-Model

Module IV: Verification, Validation & Software Testing — Topics Covered

1. Verification vs. Validation (Boehm's Criterion)
2. Black-Box Testing: Equivalence Partitioning & BVA
3. White-Box Testing: McCabe's Cyclomatic Complexity
4. Levels of Testing: Unit, Integration (Stubs/Drivers), System
5. Software Reliability: MTTF, MTBF & Availability

1. McCabe's Cyclomatic Complexity $V(G)$ Formulations

Given a Control Flow Graph (CFG) G:

1. $V(G) = E - N + 2$ (where E = edges, N = nodes)
2. $V(G) = P + 1$ (where P = predicate / decision nodes with multiple outgoing branches)
3. $V(G) = \text{Number of enclosed planar regions } R$

🏛 **BIT Mesra Exam Question (10 Marks)**

Problem: Draw the CFG and compute Cyclomatic Complexity for finding max of 3 numbers: `if (a > b) { if (a > c) max = a; else max = c; } else { if (b > c) max = b; else max = c; }`

Solution: The code contains 3 predicate/decision nodes ($P = 3$). Thus, $V(G) = P + 1 = 3 + 1 = \mathbf{4}$. There are 4 linearly independent execution basis paths requiring at least 4 test cases for 100% branch coverage.